NEXASTORE

AI Machines • Developer Workstations • Next-Generation Computing



OLLAMA + LLM

Running Ollama and Local LLMs Hardware Guide

Practical local inference planning

NexaStore Knowledge Series • 2026 Edition

1. Local Inference Is a System Workload

Running a local model is not only about downloading model weights. Model size, quantization, context length, GPU acceleration, system RAM, VRAM, storage and concurrent application services all affect the experience. Ollama-style local runtimes simplify model execution, but they do not remove hardware limits.

Development profile	System RAM	GPU VRAM direction	Storage
Learning / small models	32 GB	Integrated or 8 GB	1 TB
Regular 7B-8B local use	32-64 GB	8-12 GB	1-2 TB
Regular 13B-14B use	64 GB	12-16+ GB	2 TB+
Advanced local lab	128 GB	24-32+ GB	4 TB+
Large-model experimentation	128 GB+	High-VRAM workstation / cloud	4 TB+

2. What Consumes Memory?

Consumer	Explanation
Model weights	Quantization can substantially reduce the memory required to store/load weights.
KV cache	Grows with context length and active sequences; long contexts need additional memory.
Runtime overhead	Inference engines need working memory beyond the raw model file.
Other applications	IDE, browser, Docker, databases and vector stores compete for system memory.
Concurrency	Multiple simultaneous generations can increase memory and compute demand.

3. CPU-only vs GPU-accelerated

Mode	Advantages	Trade-offs
CPU-only	Works without discrete GPU; can use large system RAM	Usually slower token generation; high CPU utilization
GPU-resident	Fast inference when model/runtime fits VRAM	VRAM becomes a hard constraint; higher power draw
CPU/GPU split	Can run models larger than available VRAM	Performance depends on offload strategy and memory bandwidth

4. Local AI Developer Stack

Layer	Example responsibility
Local model runtime	Download, manage and serve local models
Model	Instruction, coding or embedding model
Application	.NET / Python service calling the local model endpoint
RAG layer	Chunking, embeddings, vector retrieval and context construction
Database	PostgreSQL/pgvector or another vector-capable store
Containers	Optional packaging for application services
Observability	Latency, token usage, errors, CPU/GPU and memory monitoring

5. Practical Troubleshooting

Symptom	Likely checks
Model fails to load	Available RAM/VRAM, model size, quantization, runtime version
Very slow generation	GPU acceleration, CPU offload, thermal throttling, model size
Out-of-memory during long prompts	Context length, KV cache, concurrency and available memory
System becomes unresponsive	Total RAM pressure from model + containers + IDE + databases
Disk fills unexpectedly	Downloaded model variants, caches, container images and datasets

6. Recommendation Example

If a user says, 'I use Visual Studio, Docker, PostgreSQL and want to run a 14B model locally,' the recommendation agent should combine live product specifications with this guide and the Local LLM Hardware Requirements Guide. It should favor adequate system RAM, GPU VRAM and SSD capacity rather than filtering only on an 'AI laptop' label.

Educational notice

Ollama is referenced as an example local-model runtime. This fictional NexaStore guide does not guarantee compatibility with any particular model or hardware configuration.